Richardson-seiche mixing scheme: mathematical description

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Overview

The Richardson-seiche scheme sets vertical eddy diffusivity from resolved current shear inside the surface mixed layer and from basin-scale seiche energy below it. It is selected with `diffusion_method = richardson_seiche` in `run_config.csv` and implemented in `src/jax_lakeModel_functions.py` of the Lake-M3-autodiff JAX model.

The split follows from a test on Lake Ravn (2023–2024): a 1D model resolves wind-driven currents reasonably near the surface but not below it, where they lack the basin-scale pressure gradients of a real lake. Using that deep shear in a Richardson-number closure kept the hypolimnion fully mixed. Below the surface layer, mixing therefore comes from a seiche energy budget after Goudsmit et al. (2002), converted to a diffusivity with the Osborn (1980) relation.

The other closures (`hendersonSellers` , the default, and plain `richardson`) are unchanged. All equations below describe the code as implemented, including where it departs from the cited sources.

Grid and notation

The column has N layers, numbered i = 0 at the surface to i = N − 1 at the bottom, each Δz = 0.5 m thick (`dx`), with centre depths z_i.

Symbol	Meaning	Units
A_i, A_0	Horizontal area at layer centre i; surface area	m^2
V_i, V	Layer volume (≈ A_i Δz); lake volume Σ V_i	m^3
T_i, ρ_i	Temperature; density ρ(T_i)	°C; kg m^{-3}
u_i, v_i	Eastward and northward current	m s^{-1}
E	Total basin-scale seiche energy (model state)	J

Symbol	Meaning	Units
ρ_0	Reference density: column mean in the momentum and N² terms, 1000 (`rho_fw`) in the seiche budget	kg m⁻³
W	Effective 10 m wind speed: meteo series × `WindSpeed` (`lake_config.csv`) × `wind_factor`	m s⁻¹
θ	Direction the wind blows from (meteorological convention)	degrees
I	Ice-cover indicator (1 = ice)	–
g, Δt	Gravity; model time step (3600)	m s⁻²; s

Momentum equations

Horizontal currents follow Goudsmit et al. (2002, eq. 2), with friction at the basin boundaries added:

$$\frac{\partial u}{\partial t} = \frac{1}{A}\frac{\partial}{\partial z}\left(A\,\nu\,\frac{\partial u}{\partial z}\right) + f\,v - C_{Db}\,|\mathbf{U}|\,\gamma(z)\,u$$
$$\qquad$$
$$\frac{\partial v}{\partial t} = \frac{1}{A}\frac{\partial}{\partial z}\left(A\,\nu\,\frac{\partial v}{\partial z}\right) - f\,u - C_{Db}\,|\mathbf{U}|\,\gamma(z)\,v$$

The terms are defined as follows.

- Coriolis parameter f = 2Ω sin φ, with Ω = 7.292 × 10⁻⁵ s⁻¹ and φ the lake latitude.
- Momentum viscosity ν = K_z from the mixing scheme, which assumes a turbulent Prandtl number of 1. Goudsmit et al. derive separate momentum and heat viscosities from k-ε.
- Current speed |U| = √(u² + v² + ε_U²), with ε_U = 10⁻⁴ m s⁻¹ so the derivative stays finite at rest.
- C_Db (`Cd_bottom`) is the boundary drag coefficient, 0.002 after Goudsmit et al.

The lakebed area per unit volume, γ, spreads friction by the bottom area each layer touches, as Goudsmit et al. (2002, eqs. 20–21) do for seiche energy:

$$\gamma_i = \frac{\max(A_i - A_{i+1},\,0)}{A_i\,\Delta z}\quad (i < N-1),\qquad \gamma_{N-1} = \frac{1}{\Delta z}$$

Since Σ γ_i A_i Δz = A_0, friction acts on the whole lakebed area exactly once.

Boundary conditions

At the surface, wind stress enters as a flux, switched off under ice. Lakes without a wind-direction column use a fixed θ = 270°.

$$\rho_0\,\nu\,\frac{\partial (u,v)}{\partial z}\Big|_{\mathrm{surface}} = (1-I)\,\rho_{\mathrm{air}}\,C_D\,W^2\,(-\sin\theta,\,-\cos\theta)$$

Here ρ_air = 1.225 kg m⁻³ and C_D is `Cd`. At the bottom the turbulent flux is zero; momentum is removed only by the friction term through γ_{N−1}.

Why no wind stress under ice. Under a continuous ice cover the wind acts on the ice, which on small and medium lakes is held by the shore and barely moves, so almost no momentum reaches the water. The factor (1 − I) is a model choice, not taken from Goudsmit et al. (2002), whose two lakes are effectively ice-free. It matches the rest of the model, which already sets wind-mixing energy to zero and uses a much smaller oxygen exchange rate under ice. I is binary, so partial ice cover and break-up are not represented.

An optional closed-basin constraint (`zero_net_transport` = w_z, 0 or 1) removes the area-weighted depth mean each step:

$$(u_i, v_i) \leftarrow (u_i, v_i) - w_z\,\frac{\sum_j A_j\,(u_j, v_j)}{\sum_j A_j}$$

On Ravn it worsened stratification, so w_z = 0 is recommended.

Numerical solution

Each time step is split into four parts:

1. Vertical diffusion with the wind stress as a source dt·τ/(ρ_0 Δz) in the top cell, solved with flux-form Crank–Nicolson and area-weighted faces (the same tridiagonal solver as temperature).
2. Semi-implicit friction, (u, v) ← (u, v) / (1 + Δt C_Db |U| γ_i), which preserves the current direction and is stable for any Δt.
3. Exact Coriolis rotation of (u, v) by the angle fΔt.

4. The optional zero-net-transport correction.

Surface-layer closure: gradient Richardson number

Inside the surface mixed layer, diffusivity follows a stability function of the gradient Richardson number computed from resolved shear. Buoyancy frequency and shear use forward differences, with the last layer copied from the one above:

```
N_i^2 = \frac{g}{\rho_0}\,\frac{|\rho_{i+1}-\rho_i|}{z_{i+1}-z_i},\qquad
S_i^2 = \max\!\left[\left(\frac{u_{i+1}-u_i}{\Delta z}\right)^{2} +
\left(\frac{v_{i+1}-v_i}{\Delta z}\right)^{2},\;10^{-10}\right],\qquad
Ri_i = \frac{N_i^2}{S_i^2}
```

```
K_{\mathrm{surf},i} = k_{\min} + k_{\max}\,\left(1+\alpha\,Ri_i\right)^{-n}
```

This is the stability-function form of Munk & Anderson (1948) and Pacanowski & Philander (1981). For heat, those papers correspond to (α, n) ≈ (3.33, 1.5) and (5, 3). Linear stability theory puts the critical value at Ri_c = 0.25 (Miles 1961; Howard 1961).

The parameters have distinct roles. Suppression begins near Ri ≈ 1/α, and mixing is halved at Ri = (2^(1/n) − 1)/α. For large Ri, K falls as (αRi)^(−n), so n sets how quickly mixing shuts off. Reaching k_min from k_max requires Ri_off = ((k_max/k_min)^(1/n) − 1)/α, which favours n ≈ 2.5–3.

Three implementation details matter:

- The currents used are those from the previous time step.
- |Δρ| treats unstable layering as stable; convective overturning is handled by the model's separate convection step.
- The derivative of N^2/S^2 is bounded to ±1 in the backward pass only (`_ri_safe_div`), leaving forward values unchanged. Without it, gradients reached 10^{16} or more when shear sat near its floor.

Seiche energy budget

The total seiche energy E balances wind input against friction at the basin boundaries, following Goudsmit et al. (2002, eqs. 9–15), who build on Gloor et al. (2000):

```
\frac{dE}{dt} = \underbrace{a\,A_0\,\rho_{\mathrm{air}}\,C_D\,W^3\,(1-
I)}_{P_W} \;-\; \underbrace{\gamma_s\,E^{3/2}}_{L_S},
\qquad
\gamma_s = C_{Db}\,A_0\,V^{-3/2}\,\rho_0^{-1/2}
```

- a (`seiche_alpha`) is the fraction of wind energy transferred to basin-scale seiches. Goudsmit et al. calibrated 1.7×10^{-3} for Lake Alpnach and $6.2–6.3 \times 10^{-3}$ for Lake Baldegg.
- C_D stands in for the paper's surface drag coefficient c_{10} and is the model's `Cd` .
- C_Db is the paper's effective bottom drag coefficient C_D,eff ≈ 0.002, shared with the momentum friction term.
- Wind input is zero under ice, so E then decays by friction alone.

The (1 − I) factor on P_W follows from the momentum boundary condition: seiches are excited by wind setup, the water piling up at the downwind shore, and without wind stress on the water there is no setup. Seiche energy present at freeze-up then decays by bottom friction, with a half-life of about a day on Ravn.

Time stepping

The budget is stepped semi-implicitly by splitting the loss into a factor at the old time level k and a factor solved at the new level k+1:

```
\gamma_s E^{3/2} = \underbrace{\gamma_s\sqrt{E}}_{\text{level }
k}\cdot\underbrace{E}_{\text{level } k+1}
\;\;\Rightarrow\;\;
E^{k+1}-E^{k} = \Delta t\,P_W - \Delta t\,\gamma_s\sqrt{E^{k}}\,E^{k+1}
```

Collecting the E^{k+1} terms on the left gives the update. The remaining factor E is the unknown E^{k+1}, which is why only γ_s√E^k appears in the denominator; Δt·γ_s·√E is dimensionless, as the 1 + … form requires. ε_E = 1 J keeps the derivative of √E finite at E = 0, the initial value.

```
E^{k+1} = \frac{E^{k} + \Delta t\,P_W}{1 + \Delta
t\,\gamma_s\sqrt{E^{k}+\varepsilon_E}},\qquad
L_S = \gamma_s\sqrt{E^{k}+\varepsilon_E}\;E^{k+1}
```

This form has three properties:

1. Numerator and denominator are always positive, so E stays positive for any Δt; an explicit update could overshoot through zero after a strong wind event or with a longer step.
2. It is linear in E^{k+1}, so no iteration is needed and the step is cheap to differentiate.
3. It is first-order accurate, like the rest of the model. On Ravn, Δt·γ_s·√E is about 0.03 per hourly step.

L_S is the loss rate the step actually applies, so the energy removed from the budget, Δt·L_S, is exactly the energy passed to the interior mixing and the budget closes. (An earlier version evaluated L_S as γ_s E^{k+1} √E^{k+1}, which differed by order Δt; changing it moved Mendota temperature RMSE by under 0.01 °C.)

On Ravn (2023–2024) the budget gives equivalent seiche currents of 5–11 cm/s, from √(E/(ρ_0 V)), and an energy half-life of 0.7–1.5 days. Goudsmit et al. report about 3 days for the larger Lake Alpnach.

Interior diffusivity from seiche dissipation

Below the surface layer, the seiche friction loss is spread over the lakebed and converted to a diffusivity with the Osborn relation.

The derivation follows Goudsmit et al. (2002, eqs. 20–21), whose key assumption is that seiche energy is lost by friction at the lakebed and that this loss is spread evenly over the whole bed area.

Eq. 20 gives the loss per unit lakebed area [W m⁻²]. A fraction 10√C_D,eff (about 45%, citing Imboden & Wüest 1995) is dissipated in the viscous layer at the bed without producing turbulence; the rest is divided by the total bed area:

```
P_{\mathrm{sed}} = \frac{L_S\left(1-10\sqrt{C_{D,\mathrm{eff}}}\right)}
{A_{\mathrm{boundary}}}
```

Eq. 21 turns this into production per unit mass at depth z [W kg⁻¹]. A slab between z and z + dz touches a lakebed area dA, so it receives P_sed·dA; its mass is ρ_0·A·dz:

```
P_{\mathrm{seiche}}(z) = \frac{P_{\mathrm{sed}}\,dA}{\rho_0\,A\,dz} =
\frac{1}{A}\frac{dA}{dz}\,\frac{P_{\mathrm{sed}}}{\rho_0}
```

On the model grid, (1/A)(dA/dz) becomes γ_i (the same lakebed area per unit volume used for friction), and A_boundary becomes A_0, because the side areas of all layers plus the flat bottom add up to the surface area. With C_Db for C_D,eff:

$$\varepsilon_i = \gamma_i\,\frac{L_S\,\max\!\left(1-10\sqrt{C_{Db}},\,0\right)}{A_0\,\rho_0}$$

Summed over the lake, $\Sigma\, \varepsilon_i\, \rho_0\, A_i\, \Delta z = L_S\,(1 - 10\sqrt{C_{Db}})$: all energy available for turbulence is distributed, none twice. Two approximations remain. A_0 is the horizontal projection of the bed; the true sloping area is larger by roughly $1/\cos(\text{slope})$, which Goudsmit et al. also neglect for small slopes. And Goudsmit's P_{seiche} is a production term in a TKE equation, whereas here it is used directly as the dissipation ε, assuming local balance.

Goudsmit et al. feed ε into the turbulent kinetic energy equation of a k-ε model. This model has no such equation, so it uses the Osborn (1980) relation, capped at k_{max} and with N^2 floored for well-mixed water:

$$K_{\mathrm{int},i} = \min\!\left(\frac{\Gamma\,\varepsilon_i}{\max\!\left(N_i^2,\,10^{-7}\right)},\;k_{\max}\right)$$

Γ (`mixing_efficiency`) is 0.2, the upper bound proposed by Osborn. The relation assumes local balance between turbulence production and dissipation, with no memory and no vertical spreading of turbulence. Wüest et al. (2000) use a comparable energy-balance approach to estimate lake diffusivities.

Combining surface and interior mixing

A smooth density-based weight marks the surface mixed layer, and the final diffusivity blends the two closures. With σ the logistic function and ρ_{surf} the density of the top layer:

$$w_i = \sigma\!\left(\frac{\rho_{\mathrm{surf}} + \Delta\rho_c - \rho_i}{\delta\rho}\right),\qquad K_{z,i} = \max\!\left(w_i\,K_{\mathrm{surf},i} + (1-w_i)\,K_{\mathrm{int},i},\;k_{\min}\right)$$

$\Delta\rho_c$ (`sbl_drho` , 0.05 kg m^{-3}) sets the density step that defines the base of the mixed layer, and $\delta\rho$ (`sbl_width` , 0.01 kg m^{-3}) sets how sharp the transition is. A sigmoid rather than a hard threshold keeps the mixed-layer depth differentiable for calibration and machine-learning training.

K_z is used for diffusion of temperature, O_2 and dissolved organic carbon (Crank–Nicolson), for particulate organic carbon settling, and as the momentum viscosity ν.

Order within one model step

The sequence follows `full_step`:

1. Surface heat fluxes and ice.
2. Wind mixing of the surface layer.
3. Seiche energy update and K_z, using the previous step's currents.
4. Diffusion of temperature and dissolved substances, and particle settling.
5. Convective overturning.
6. Water-quality reactions and outflow.
7. Momentum step using this step's K_z.

Parameters

All parameters are read from `model_params.csv`; those marked calibratable can be passed to `calibrate_M3_jax.py --params`.

Symbol	`model_params.csv` name	Default	Recommended for Ravn	Calibratable	Source
k_min	`k_min`	1.4×10^{-7} $m^2\ s^{-1}$	1.4×10^{-7} $m^2\ s^{-1}$	yes (wide steps)	molecular limit
k_max	`k_max`	1×10^{-2} $m^2\ s^{-1}$	3×10^{-4} $m^2\ s^{-1}$	yes (wide steps)	turbulent limit
α	`alpha`	10	10	yes	Munk & Anderson 1948; Pacanowski & Philander 1981
n	`n`	1.5	3 (fixed)	possible, not advised	as above
a	`seiche_alpha`	2×10^{-3}	2×10^{-5}	yes (wide steps)	Goudsmit et al. 2002, Table 1
C_Db	`Cd_bottom`	0.002	0.002	yes	Goudsmit et al. 2002
Γ	`mixing_efficiency`	0.2	0.2 (fixed)	no	Osborn 1980
Δρ_c	`sbl_drho`	0.05 kg m^{-3}	0.05 kg m^{-3}	yes	model choice
δρ	`sbl_width`	0.01 kg m^{-3}	0.01 kg m^{-3}	no	model choice
C_D	`Cd`	0.0013	as calibrated	yes	shared with heat-flux and wind-mixing code

Symbol	`model_params.csv` name	Default	Recommended for Ravn	Calibratable	Source
w_z	`zero_net_transport`	1	0	no	model choice

In steady state ε scales with a, so K_int depends only on the product Γa and the two cannot be calibrated separately; Γ is therefore fixed. On Ravn, a ≈ 2×10^{-5} fits far better than the paper's 10^{-3} range. That suggests the Osborn shortcut with Γ = 0.2 overstates mixing relative to a full k-ε model, so a is best read as an effective calibration parameter rather than a literal energy fraction.

Differences from Goudsmit et al. (2002) and limitations

The scheme borrows Goudsmit's momentum equations and seiche budget but replaces their k-ε turbulence model with simpler algebraic closures.

- **No k-ε model.** There is no turbulent kinetic energy equation and no buoyancy production or destruction term. Mixing comes from an algebraic Richardson-number function near the surface and the Osborn relation below it.
- **No distribution parameter q.** The paper calibrates a parameter q (0.77–1.02) in the distribution of seiche-generated mixing; it is not included here.
- **Friction on mean currents.** Boundary friction acts directly on the resolved currents, which the paper's eq. 2 does not include; there, friction only drains the seiche energy.
- **Surface layer by density threshold.** The mixed layer is identified from a smooth density threshold, not from the turbulence model.
- **Turbulent Prandtl number of 1.** Momentum and heat share one diffusivity.
- **Single water column.** Only the barotropic (depth-uniform) pressure gradient could be represented, and it was found to worsen stratification; baroclinic return flows above a tilting thermocline cannot be resolved. Surface currents remain about 0.2 m s^{-1} but no longer drive deep mixing.
- **Boundary mixing as a basin average.** Interior mixing is the area average of near-bed mixing and ignores internal-wave breaking in open water.

On Ravn (2023–2024, uncalibrated, a = 2×10^{-5}, k_max = 3×10^{-4}, n = 3), the scheme gave a temperature RMSE of 2.82 °C, against 3.61 °C for Henderson-Sellers with the same `model_params.csv`. It has not yet been tested on Lake Mendota.

References

Only Goudsmit et al. (2002) was read in full for this description; the remaining entries are from memory and should be checked before citing.

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